

Mariana Rius

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EDUCATION

Stony Brook University, School of Marine and Atmospheric Sciences

Doctor of Philosophy, Biological Oceanography

Stony Brook, NY

August 2014 – January 2021

College of Charleston, Department of Biology

Bachelor of Science, Marine Biology

Charleston, SC

August 2009 - May 2012

Minors, Chemistry & Spanish

DOCTORAL AND POSTDOCTORAL RESEARCH

Stony Brook University, School of Marine and Atmospheric Sciences, Department of Ecology and Evolution, Principal Investigators: Dr. Jackie Collier and Dr. Joshua Rest, *Stony Brook, NY*

February 2021 - August 2021. Differential gene expression (RNAseq) in pigmented (wildtype) and genetically modified carotenoid-less *Aurantiochytrium limacinum* (heterotrophic marine protist, Stramenopile) strains under nutrient and oxidative stress growth conditions.

August 2014 - January 2021. Determining the evolutionary origins through phylogenetics of thraustochytrid carotenoid biosynthesis; Developing and applying a transformation system for the inactivation of carotenoid biosynthesis in *Aurantiochytrium limacinum*.

TEACHING EXPERIENCE

Stony Brook University, *Stony Brook, NY*

Guest Lecturer - Bioinformatics and Computational Biology (BIO 312), College of Arts and Sciences
Phylogenetics as a Research Application, September 30, 2019

Guest Lecturer - Oceanography (MAR 104), School of Marine and Atmospheric Sciences
Marine Primary Producers, November 8, 2016
Ocean Acidification, December 6, 2016

Instructor - Fundamentals of Scientific Inquiry in Biological Sciences IIA (BIO 205)
Undergraduate Biology, College of Arts and Sciences, January 2015 - May 2015

Teaching Assistant - Marine Mammals (MAR 370)
School of Marine and Atmospheric Sciences, August 2014 - Dec 2014

College of Charleston, *Charleston, SC*

Instructor (Peer Teacher) - Spanish 202 Conversation (SPAN202-C)
Department of Hispanic Studies January 2011 – May 2012

High Schools

Guest Lecturer - AP Spanish, Port Jefferson High School, *Port Jefferson, NY*
Acidificación del Océano, December 12, 2016

Guest Lecturer - The Frost School, *Rockville, MD*
Molecular and Marine Biology, November 23, 2016

LEADERSHIP AND OUTREACH

The Carpentries (<https://carpentries.org/>)

Certified Instructor, Instructor Training Participant

Online

June – July 2021

Stony Brook University, Women in Science and Engineering

Leadership Workshop Series Participant

Stony Brook, NY

March 2021 - April 2021

Stony Brook University, Graduate Writing Programs

Dissertation/ Thesis Writing Boot Camp Program Leader

Stony Brook, NY

Feb 2021 - May 2021

Stony Brook University, Collier and Rest Labs

Phylogenetics Pipeline Tutorial Leader

Stony Brook, NY

June 2020

Growthcurver R package Workshop Leader	April 2020
Organizer and Moderator of the 2019 Summer Research Symposium	August 2019
<u>Protocols.io</u>	<i>protocols.io</i>
EBSCO Webinar Panelist, Supporting Open Research Workflows and Digital Collaboration	May 4, 2020
Ambassador, Ambassador Program	December 2018 - Present
<u>Long Island Aquarium</u>	<i>Riverhead, NY</i>
Meet an Oceanographer - Microorganisms	May 20, 2017
<u>Stony Brook University, Center for Inclusive Education</u>	<i>Stony Brook, NY</i>
Panelist Speaker for Graduate School Preview Day	August 18, 2017
Panelist Speaker for Graduate School Preview Day	November 17, 2016
Diversity Recruitment Team at SACNAS 2016, Long Beach, CA	October 2016
Diversity Recruitment Team at SACNAS 2015, Washington D.C.	October 2015
<u>Mental Health Association, Families Foremost Support Center</u>	<i>Rockville, MD</i>
Volunteer Adult Education Tutor	September 2013 – June 2014
<u>College of Charleston, Counseling and Substance Abuse Services</u>	<i>Charleston, SC</i>
Volunteer Peer Counselor	August 2010 – May 2012

MENTORING EXPERIENCE

Department of Chemistry, Stony Brook University, <i>Stony Brook, NY</i>	
May 2019 - August 2021. Michael Horowitz, Masters Student: Are carotenoids involved in the oxidative stress response of <i>Aurantiochytrium limacinum</i> ?	
Department of Ecology and Evolution, Stony Brook University, <i>Stony Brook, NY</i>	
November 2018 - May 2020. Xegfred Lou Quidet, Undergraduate Student: The effects of hydrogen peroxide on the growth of <i>Aurantiochytrium limacinum</i> .	
November 2018 - May 2019. Bingyuan Zhu, Undergraduate Student: The relationship between cell density and optical density (OD) of <i>Aurantiochytrium limacinum</i> .	
June - September 2019. Elizabeth Yaboni, Summer High School Student: An investigation of the biological purpose of carotenogenesis and a method for assessing viability in <i>Aurantiochytrium limacinum</i>	
Simons Summer Research Program, Stony Brook University, <i>Stony Brook, NY</i>	
June - August 2018. Sarah Kelso, Simons Summer High School Student: Inactivation of the carotenoid synthesis gene in a non-photosynthetic marine protist	
June - August 2018. Kylen Bao, Simons Summer High School Student: Inactivation of a proteorhodopsin-like gene in <i>Aurantiochytrium limacinum</i> by double homologous recombination	
School of Marine and Atmospheric Sciences, Stony Brook University, <i>Stony Brook, NY</i>	
May - August 2015. Kevin Crawford, Undergraduate Student: Effects of varying media glucose:nitrogen ratio on growth of <i>Aurantiochytrium limacinum</i>	

PROFESSIONAL DEVELOPMENT

<u>Stony Brook University</u> , The Graduate School	<i>Stony Brook, NY</i>
Ph.D. Career Ladder Program Participant	May 2020 - August 2020
<u>SUNY Research Foundation</u>	<i>Albany, NY</i>
Mentoring Program Participant - Protégé	Sept 2019 - June 2020

PUBLICATIONS

- Rius, M.**, Rest, J.S., Filloramo, G.V., Novák Vanclová, A.M.G., Archibald, J.M., Collier, J.L. (2021). A novel origin of eukaryotic carotenoid production in a diverse group of heterotrophic protists. (in preparation)
- Rius, M.** (2021). *Evolutionary origins, regulation, and function of carotenoid biosynthesis in the marine heterotrophic eukaryote, Aurantiochytrium limacinum* (Publication No. 28315974) [Doctoral dissertation, Stony Brook University]. ProQuest Dissertations Publishing.
- Faktorová, D., et al. (2020). Genetic tool development in marine protists: emerging model organisms for experimental cell biology. *Nat Methods* 17, 481–494.
- Rius, M.**, Collier, J.L. (2019). Electroporation of *Aurantiochytrium limacinum* (ATCC MYA-1381). protocols.io <https://dx.doi.org/10.17504/protocols.io.8xyhxpw>.
- Rius, M.** (2018). Zoospore Isolation of *Aurantiochytrium limacinum*. protocols.io <https://dx.doi.org/10.17504/protocols.io.u73ezqn>.
- George A., Zand D.J., Hufnagel R.B., Sharma R., Sergeev Y.V., Legare J.M., Rice G.M., Scott Schwoerer J.A., **Rius M.**, Tetri L. Gamm D.M., Bharti K., Brooks B.P. (2016). Biallelic Mutations in MITF Cause Coloboma, Osteopetrosis, Microphthalmia, Macrocephaly, Albinism, and Deafness. *Am J Hum Genet.* Dec 1;99(6):1388-1394.

FELLOWSHIPS AND AWARDS

<u>Stony Brook University, Center for Inclusive Education</u>	<i>Stony Brook, NY</i>
Turner Summer Research Grant (\$5000)	2020
Turner Conference Travel Award (\$1200)	2016
Turner Conference Travel Award (\$695)	2015
Dr. W. Burghardt Turner Fellowship (\$50,000)	2014

RELEVANT SKILLS

Languages: Native fluency in Spanish and English; Beginner French and Italian

Bioinformatics/ Programming: Proficiency in R/Rstudio, Unix/BASH, RNA-seq data analysis, phylogenetics, MobaXTerm, High Performance Computing Clusters (module, slurm), AWS Console, beginner python

Computer Skills: Proficiency in Microsoft Word, PowerPoint, Excel, Outlook, BioEdit, Mega, Dendroscope, Notung, Inkscape, ZEN 3.0, i-control, Endnote, Google Drive, Slack, Zoom, protocols.io, beginner LaTeX

Technical Lab Bench Skills: primer design, Infusion cloning, *E. coli* transformation including plasmid extraction and cryopreservation, genomic DNA extraction, RNA extraction, nucleic acid quantification (NanoDrop), PCR and colony PCR (*E. coli* and *A. limacinum*), gel electrophoresis, gel extraction, restriction digest, microplate growth assays including nutrient optimization, antibiotic inhibition, and growth with oxidative stressors, thraustochytrid culture (liquid and solid media) and transformation by electroporation, fluorescence microscopy, spectrophotometry, Nile red and propidium iodide staining